LESSON 8.2

Write and Solve Equations Using Addition or Subtraction

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve one-step addition and subtraction equations using inverse operations.

WHY THIS MATTERS Detective Chen found a coded message: 'The suspect's locker number plus 23 equals 58.' She needs to figure out the locker number to crack the case. The equation is n + 23 = 58. What is n?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Equation *Spanish: Ecuación*	A math sentence with an equal sign showing both sides are the same.	x + 25 = 60 — both sides equal 60 when x = 35
Inverse operation *Spanish: Operación inversa*	Two math actions that undo each other, like × and ÷.	+ and −
Isolate *Spanish: Despejar*	To get the letter by itself on one side.	get x by itself
Solution *Spanish: Solución*	A number that makes the equation or inequality true.	x = 7 is the solution to x + 5 = 12 because 7 + 5 = 12 ✓
One-step equation *Spanish: Ecuación de un paso*	An equation you can solve in a single step using one inverse operation.	x + 7 = 20
Balance *Spanish: Equilibrio*	Keeping both sides of an equation equal by doing the same thing to each side.	do the same to each side

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A team had some points, then scored 14 more and has 40. Solve y + 14 = 40.

1. Subtract 14 from both sides: y + 14 − 14 = 40 − 14.
2. Simplify: y = 26.
3. Check: 26 + 14 = 40.

Answer: y = 26 points

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Solve x − 4 = 9. Is x being added or subtracted?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The variable has _____ added to it, so I _____ from both sides.
- I must do the same thing to both sides because _____ .
- My solution is _____ , and my check shows _____ .
- Addition and subtraction are inverse operations because _____ .

Word bank equation | variable | inverse operation | subtract | isolate | balance | both sides | solution | check | undo | Property of Equality

WATCH OUT!

A common mistake

A common mistake is applying the same operation instead of the inverse

LESSON 8.2

Write and Solve Equations Using Addition or Subtraction

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve one-step addition and subtraction equations using inverse operations.

WHY THIS MATTERS Detective Chen found a coded message: 'The suspect's locker number plus 23 equals 58.' She needs to figure out the locker number to crack the case. The equation is n + 23 = 58. What is n?

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A team had some points, then scored 14 more and has 40. Solve y + 14 = 40.

1. Subtract 14 from both sides: y + 14 − 14 = 40 − 14.
2. Simplify: y = 26.
3. Check: 26 + 14 = 40.

Answer: y = 26 points

1 **Circle the letter of the best answer. Show how you know.**

Solve: x + 9 = 15

(A) x = 6
(B) x = 9
(C) x = 15
(D) x = 24

STEPS TO TRY
1. x has 9 added to it. To get x alone, you need to UNDO that addition.
2. Use the inverse operation: subtraction undoes addition. Whatever you do, do it to BOTH sides to keep the balance.
3. Subtract 9 from both sides: x = 15 − 9. Then check by adding 9 back to your result — do you land on 15?

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Solve: $y - 7 = 20$

(A) $y = 7$

(B) $y = 13$

(C) $y = 27$

(D) $y = 140$

STEPS TO TRY

1. Look at what's happening to y: 7 is being SUBTRACTED. Which operation undoes subtraction?
2. Add the same amount to both sides so the equation stays balanced and y ends up alone.
3. Add 7 to both sides: $y = 20 + 7$. Check your result: does it minus 7 give back 20?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Solve: $n + 15 = 42$

(A) $n = 15$

(B) $n = 27$

(C) $n = 42$

(D) $n = 57$

STEPS TO TRY

1. n is being increased by 15. To find n alone, you must undo that addition.
2. Subtract 15 from both sides — the +15 and −15 cancel on the left, leaving just n.
3. Compute $42 - 15$ carefully ($42 - 10 = 32$, then take away 5 more). Check by adding 15 back to your answer.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Solve: m − 11 = 25

(A) m = 14

(B) m = 25

(C) m = 36

(D) m = 275

STEPS TO TRY

1. m has 11 taken away from it. Think about which operation puts that 11 back.
2. Addition undoes subtraction: add 11 to both sides to leave m by itself.
3. Compute 25 + 11 for m. Then check: your answer minus 11 should equal 25.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Why do you have to subtract 19 from BOTH sides, not just one?

SENTENCE STARTER I subtract from both sides because an equation is like a ___ — to keep it balanced, any operation on one side must ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help